PARKINSON'S EARLY DETECTION USING CNN & VGG16 DEEP LEARNING MODELS

Palla Saikeerthana , Madira Gowtham , Kotra Manjusha , Juturu Sai Vivek

Department of Artificial Intelligence & Data Science, Vasireddy Venkatadri Institute of Technology, Guntur , Andhra Pradesh, India

Guide: Dr . S . Ravindra , Professor

ABSTRACT

Early detection of Parkinson's disease remains a significant challenge in healthcare due to the gradual onset of symptoms and reliance on clinical observation. Traditional diagnostic methods are often subjective, time-consuming, and may fail to identify the disease at its early stages. To address this limitation, the proposed AI-based Parkinson's Early Detection System aims to provide a non-invasive, efficient, and automated screening solution using image analysis techniques. The system leverages advanced deep learning models, such as Convolutional Neural Networks (CNN) and VGG16, to analyze spiral and wave drawing patterns that reflect fine motor control impairments associated with Parkinson's disease. By capturing subtle irregularities in handwriting, including tremors, distortion, and micrographia, the model is able to identify early indicators of the disease with improved accuracy. The platform allows users to upload drawing samples through a user-friendly interface, where images are preprocessed and analyzed in real time. The system generates prediction results along with confidence scores and detailed feature analysis, providing meaningful insights into the user's condition. By utilizing machine learning and image processing techniques, this project offers a cost- effective and accessible solution for early Parkinson's screening.

*I.*INTRODUCTION

Parkinson's disease is a progressive neurological disorder that affects movement and motor coordination, making early detection critical for effective management and treatment. Traditional diagnostic approaches primarily rely on clinical observation and specialist evaluation, which can be time-consuming, subjective, and often identify the disease only at later stages. Additionally, these methods may not effectively capture subtle early symptoms, such as minor handwriting irregularities and fine motor impairments.

To address these challenges, the proposed Parkinson's Early Detection System aims to provide an efficient, non-invasive, and automated screening solution using image analysis techniques. The system leverages advanced deep learning models, including Convolutional Neural Networks (CNN) and VGG16, to analyze spiral and wave drawing patterns that reflect motor control abnormalities associated with Parkinson's disease. The platform allows users to upload drawing samples through a user-friendly interface, where images are preprocessed and analyzed in real time. The system identifies key features such as tremors, distortion, line irregularities, and micrographia to generate prediction results along with confidence scores and detailed analysis. By automating the detection process, the system reduces the need for manual evaluation and enhances accuracy and efficiency.

II. LITERATURE REVIEW

Parkinson's disease detection has gained significant attention with advancements in artificial intelligence (AI) and medical image analysis. Traditional diagnostic methods rely heavily on clinical expertise and often fail to detect the disease at early stages. Recent research focuses on AI-driven approaches that analyze various data types such as voice signals, handwriting patterns, and medical imaging to improve early detection accuracy. Several studies utilize spiral and wave drawing analysis to detect motor impairments. These systems analyze features such as tremors, line irregularities, and distortions using machine learning algorithms. While effective, many models depend on handcrafted features and may lack generalization across diverse datasets.Deep learning models such as Convolutional Neural Networks (CNN) have shown promising results in medical diagnostics. These models automatically extract relevant features from images, improving detection accuracy. However, they require large datasets and high computational resources for training.Pre-trained models like VGG16 are widely used to enhance performance when

datasets are limited. These models leverage previously learned features to improve classification accuracy. Despite their advantages, they may require fine-tuning for specific medical applications.

III. PROPOSED SYSTEM

The Parkinson's Early Detection System is designed to provide an intuitive and automated screening experience using image analysis. The system allows users to interact through a simple web interface where they can provide basic details and upload spiral and wave drawing samples for analysis. Unlike traditional systems, this platform focuses on non- invasive detection using handwriting patterns. The process begins when the user accesses the application and navigates to the assessment section. Users are required to enter essential details such as age, gender, and dominant hand, followed by uploading drawing images. Once the inputs are provided, the system processes the data and sends it to the deep learning model for prediction. Fig: System Architecture After submitting the images, the system performs preprocessing and analysis using trained models such as CNN and VGG16. The results are then generated in real time, including risk level and confidence score. Users can view the results immediately and understand their condition through detailed analysis. The system also stores prediction results for future reference.

System Architecture

The Parkinson's Disease Detection System follows a client-server architecture built using modern web technologies. The frontend is developed with React 18 and TypeScript, utilizing Vite as the build tool and Tailwind CSS for responsive styling. The user interface captures patient demographics and accepts spiral and wave drawing images through a multi-step assessment workflow. Upon submission, images are transmitted to a Flask-based backend server via REST API endpoints. The backend preprocesses the uploaded images—resizing them to 224×224 pixels and normalizing pixel values—before feeding them into a fine-tuned VGG16 convolutional neural network for inference. The model returns prediction scores, which the backend aggregates into an overall risk assessment along with detailed tremor, micrographia, and consistency metrics. Results are rendered on the frontend through interactive charts built with Recharts, and users can export comprehensive clinical reports as downloadable PDF documents using jsPDF.

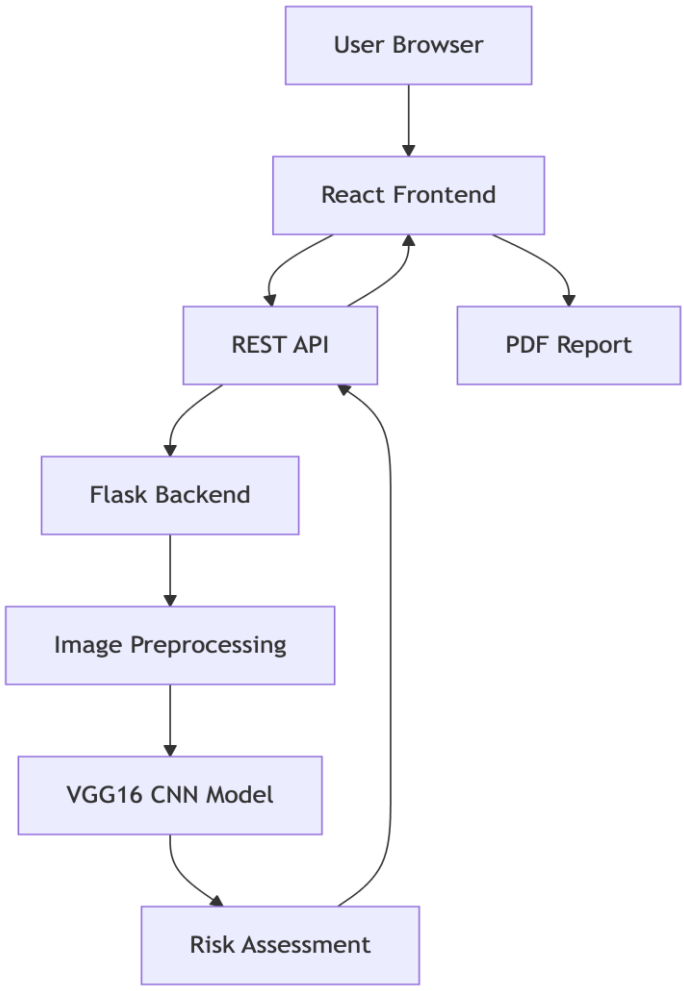

Fig: System Architecture

Block Diagram

The block diagram of the Parkinson's Early Detection System illustrates the overall workflow and interaction between the main components, including the frontend interface, backend processing unit, image preprocessing module, and deep learning model. The process begins with the user interface, where users upload their drawing images. These images are then sent to the backend server for further processing. In the backend, an image preprocessing module is applied to enhance the input data. This includes operations such as resizing, grayscale conversion, and noise removal to ensure consistency and improve model performance. The preprocessed images are then passed to the deep learning model, which performs feature extraction and classification to detect signs of Parkinson's disease. Finally, the prediction results are generated and sent back to the frontend, where they are displayed to the user in a clear and understandable format. This structured workflow ensures smooth data flow, efficient

processing, and accurate prediction of results.

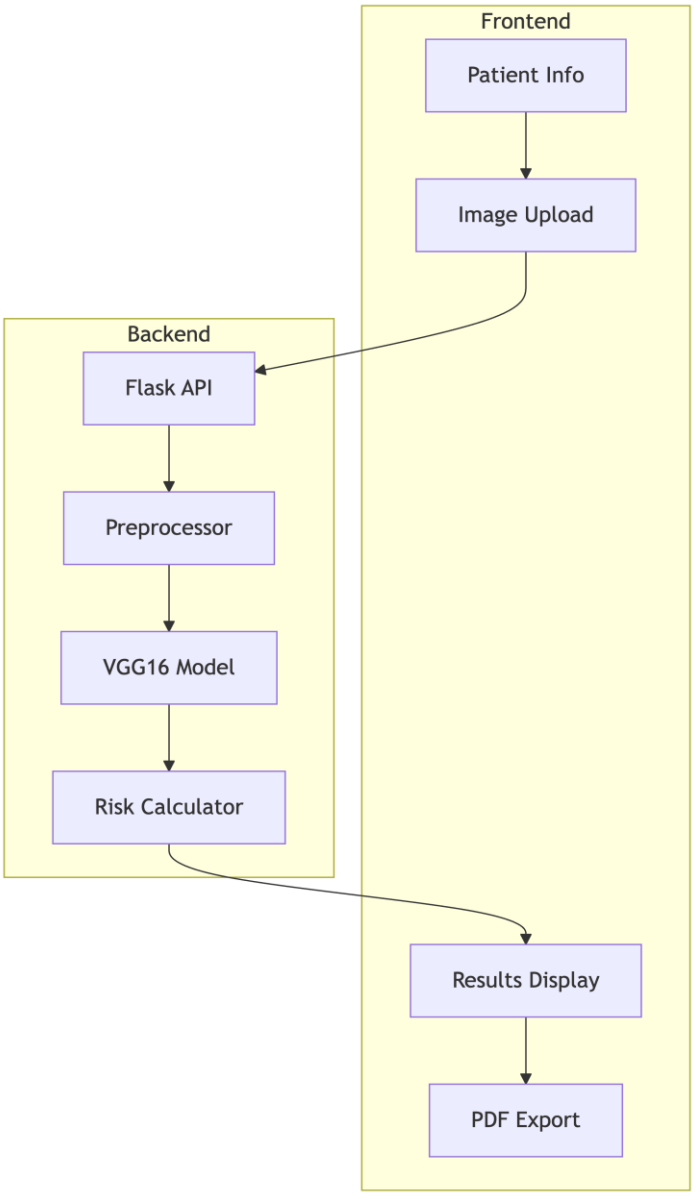

Fig: Block Diagram

Class Diagram

The class diagram represents the core components of the Parkinson's Early Detection System and illustrates the relationships between different modules. The Image class is responsible for handling uploaded drawing samples, including attributes such as image ID and file path, and methods for uploading and preprocessing images before analysis. The Prediction class stores the output generated by the system, containing details such as prediction ID, result, and confidence score, which represent the outcome of the analysis. The Model class represents the deep learning component, such as CNN or VGG16, which processes the preprocessed images and generates predictions using functions like loading the model and performing classification. The Database class, which is optional in implementation, is used to manage storage operations such as saving images and prediction results and retrieving past data if required. Overall, the interaction between these classes ensures efficient data flow, modular design, and scalability of the system.

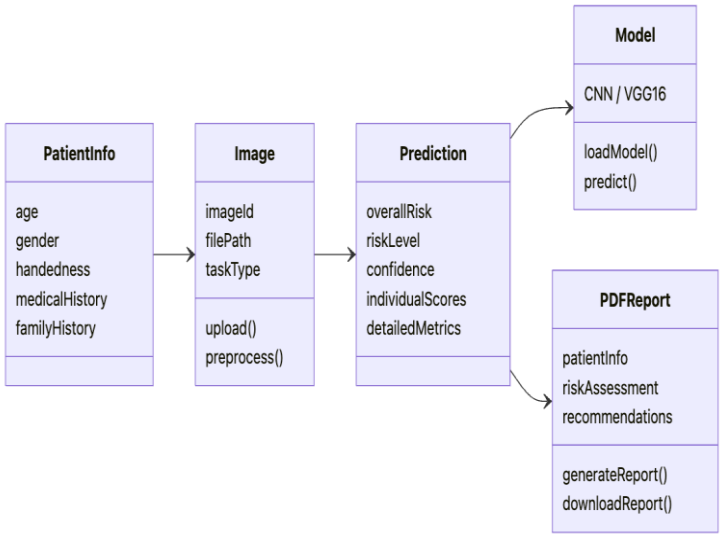

Fig: Class Diagram

Sequence Diagram

The sequence diagram illustrates the interaction between the User, Web Interface, Backend Server, Deep Learning Model, and system components during the prediction process. The process begins when the user accesses the application and uploads drawing images through the web interface. The interface then forwards the uploaded images to the backend server, where validation and preprocessing operations such as resizing and noise removal are performed. After preprocessing, the data is sent to the deep learning model, which analyzes the images and generates prediction results based on learned patterns. The model returns the results to the backend server, which processes the output and sends it back to the frontend interface. Finally, the results are displayed to the user in a clear format, including the predicted condition and confidence score. This sequence ensures smooth communication between system components, efficient data processing, and real-time prediction of Parkinson's disease.

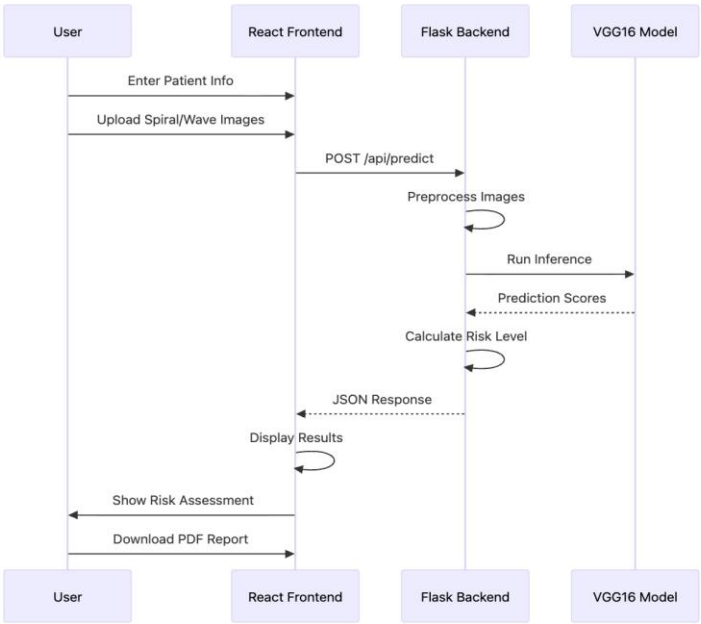

Fig: Sequence Diagram

IV. METHODOLOGY

The development of the Parkinson's Early Detection System follows a structured methodology that integrates image processing and deep learning techniques to achieve accurate and efficient prediction. The system is designed to provide a non- invasive screening approach by analyzing spiral and wave drawing patterns.

1. **Problem Definition**

   The primary objective of the system is to detect early signs of Parkinson's disease using handwriting analysis.Traditional diagnostic methods are often subjective and time-consuming, creating a need for an automated and accessible solution.

2. **User Input & Data Collection**

   Users provide basic details such as age, gender, and dominant hand through a simple interface. Additionally, users are required to draw spiral and wave patterns on paper and upload the images foranalysis. These inputs serve as the primary datafor detecting motor impairments.

3. **Image Preprocessing**

   The uploaded images undergo preprocessing steps to improve quality and consistency. This includes resizing images to a standard dimension, converting them to grayscale, removing noise using filtering techniques, and normalizing pixel values. These steps ensure that the data is suitable for deep learning models.

4. **Feature Extraction & Analysis**

   The system analyzes important features from the images, such as line waviness, tremor patterns, spiral distortions, and writing consistency. These features are critical indicators of Parkinson's disease and are extracted automatically using deep learning techniques.

5. **Deep Learning Model Implementation**

   Convolutional Neural Networks (CNN) and pre-trained models like VGG16 are used to classify the input images. The models are trained on labeled datasets to distinguish between healthy and Parkinson's-affected patterns. The system processes the input images and generates prediction results along with confidence scores.

6. **Result Generation & Visualization**

   The output of the model is displayed to the user in a clear and understandable format. The system provides risk levels (low, medium, high), confidence scores, and detailed analysis of each parameter. This helps users interpret their results effectively**.**

7. ***Results Export & Reporting***

   Results are generated in real-time and can be exported as PDF reports for future reference. This allows users to access previous analyses and track their condition over time. This methodology ensures an efficient workflow from data collection to prediction, enabling accurate, real-time, and scalable Parkinson's disease detection using artificial intelligence.

V. RESULTS

The developed Parkinson's Early Detection System provides an intuitive user interface and accurate prediction results through AI-based image analysis. The system is designed to ensure ease of use while delivering reliable screening outcomes in real time.

1. **User Interface**

   The Home Page serves as the main entry point of the application, providing users with an overview of the system and its features. It includes a navigation bar with sections such as Home, About, How It Works, Assessment, and Results. The page also contains a clear call-to-action button

that allows users to start the assessment process easily.

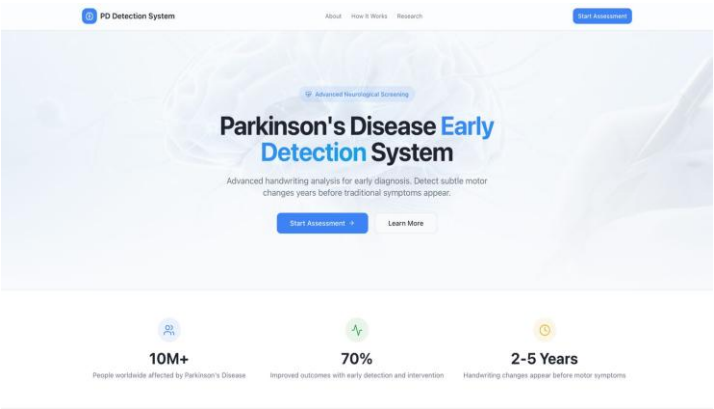

Fig: Landing Page

The About Page provides detailed information about Parkinson's disease, its symptoms, and the importance of early detection. It explains how the system uses AI and handwriting analysis to identify early signs of the disease. The page is designed with a simple and informative layout to improve user understanding.

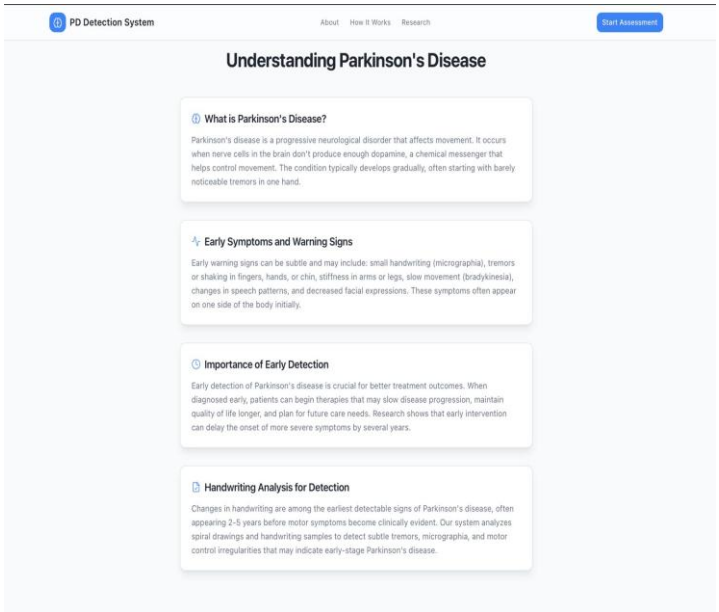

Fig: About Section

The How It Works Page explains the step- by-step process of the system. It guides users through drawing spiral and wave patterns, uploading images, and receiving prediction results. This helps users clearly understand the workflow of the system.

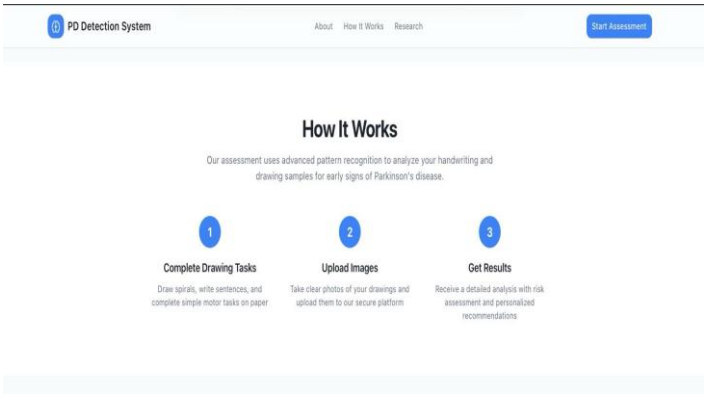

Fig: How It Works Section

The Assessment Page is the core functionality of the system, where users enter their details and upload drawing samples. Users provide basic information such as age, gender, and dominant hand, followed by uploading spiral and wave images required for analysis.

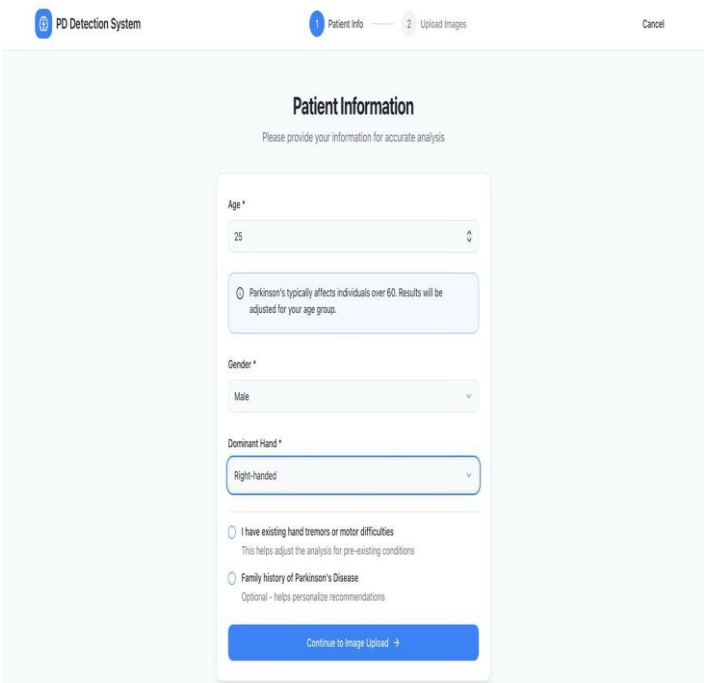

Fig: Assessment Page

2. Image Upload & Processing

The system allows users to upload multiple drawing samples, including spiral and wave patterns. Once uploaded, the system validates the images and prepares them for preprocessing. This ensures that only valid inputs are processed by the model.

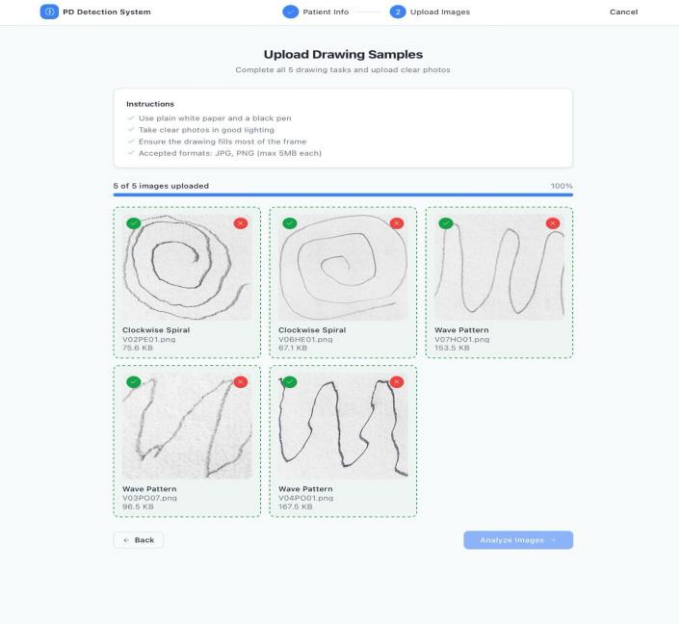

Fig: Upload Drawing Samples

3. Prediction Results

The system generates prediction results based on deep learning analysis of the uploaded images. The results include a risk classification (low, medium, or high) along with a confidence score indicating prediction reliability.

A detailed breakdown of parameters such as tremor detection, line waviness, spiral consistency, and micrographia is also provided. Additionally, individual image analysis is displayed to give users a clear understanding of each input.

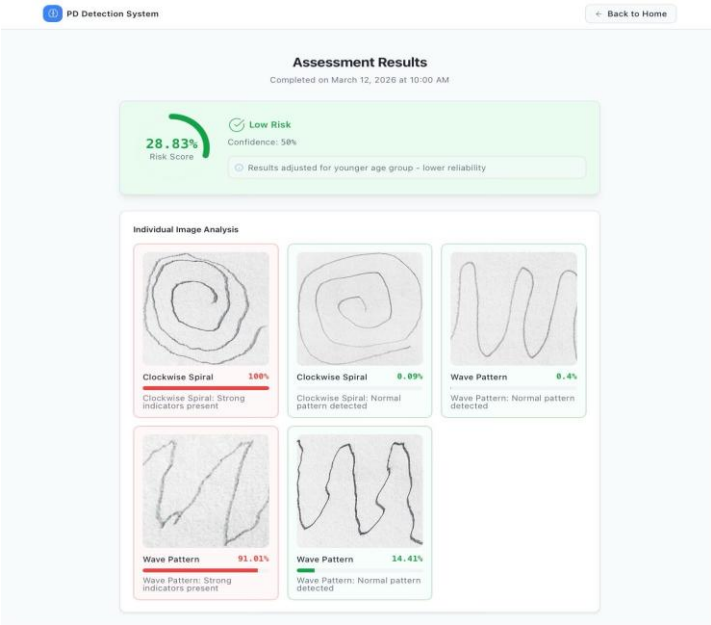

Fig: Results Page

4.System Performance

The system demonstrates efficient performance with real-time processing and accurate predictions. The integration of image preprocessing and deep learning models ensures reliable analysis of handwriting patterns. The platform provides quick response times, making it suitable for practical screening applications.

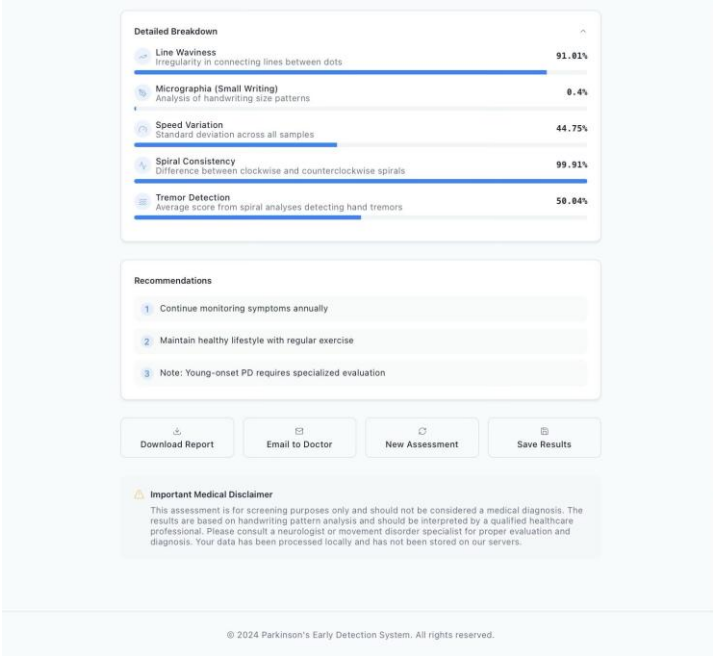

Fig: System Performance Section

VI. CONCLUSION

The Parkinson's Early Detection System successfully demonstrates the application of artificial intelligence in medical diagnostics. By utilizing deep learning models such as CNN and VGG16, the system provides an efficient and non-invasive approach to detect early signs of Parkinson's disease.

The platform offers a user-friendly interface for uploading drawing samples and receiving real- time predictions. The integration of multiple modules, including image processing, AI-based prediction, and database storage, ensures smooth system functionality and reliable results.

Overall, this system provides a scalable and cost-effective solution for early Parkinson's screening, supporting both users and healthcare professionals. It highlights the potential of AI in improving early diagnosis and enhancing healthcare accessibility.

REFERENCES

1. M. J. Farrer, "Genetics of Parkinson disease: paradigm shifts and prospects," ***Nature Reviews Genetics***, vol. 7, no. 4, pp. 306–318, 2006.
2. M. Lozano and A. E. Lang, "Parkinson's disease," ***New England Journal of Medicine***, vol. 339, no. 15, pp. 1044–1053, 1998.
3. Cantürk and F. Karabiber, "A machine learning system for the diagnosis of Parkinson's disease from speech signals," ***Arabian Journal for Science and Engineering***, vol. 41, pp. 5049–5059, 2016.
4. Pereira et al., "HandPD Dataset," 2017. [Online]. Available: https://www.ufpe.br/handpd
5. K. Mader, "Parkinson's Drawings Dataset," Kaggle, 2019. [Online]. Available: https://www. ka ggle. c om/ kma der /pa r kinsons - drawings
6. Huseyn, "Deep Learning Based Early Diagnosis of Parkinson's Disease," arXiv, Cornell University, 2020.
7. Editorial Team, "Diagnosis: Early Symptoms & Early Diagnosis," ParkinsonsDisease.net, 2017.
8. Pereira, "Convolutional Neural Networks applied for Parkinson's disease identification," ***Machine Learning for Health Informatics***, 2017.
9. C. Pereira, "Deep Learning-Aided Parkinson's Disease Diagnosis from Handwritten Dynamics," ***Graphics, Patterns and Images (SIBGRAPI)***, IEEE Conference, 2016.
10. M. Gil-Martin, "Parkinson's Disease Detection from Drawing Movements using CNN," ***Electronics (MDPI)***, 2019.
11. Dragana Miljkovic et al., "Machine Learning and Data Mining Methods for Managing Parkinson's Disease," LNCS 9605, pp. 209–220,2016.
12. International Conference on Advanced Computation and Telecommunication (ICACAT), Bhopal, India, 2018.